

Dominic DiSanto (he/him)

jdisanto@g.harvard.edu | <https://domdisanto.github.io/>

EDUCATION

Harvard University

August 2022 - Present

T.H. Chan School of Public Health

Doctoral Candidate in Biostatistics

Dissertation: *Inferential Frameworks for Biomedical Decision Processes*

Advised by Junwei Lu and Tianxi Cai

Cognate in History & Philosophy of Science

University of Pittsburgh

Graduate School of Public Health

August 2019 - Dec 2020

M.Sc. in Biostatistics, Concentration in Health Data Science

Thesis: *Treelet Dimension Reduction of Diagnoses Among Intensive Care Unit Admissions*

Dietrich School of Arts & Sciences

August 2014 - May 2018

B.Sc. in Neuroscience

PROFESSIONAL EXPERIENCE

Statistician

June 2020 - July 2022

U.S. Department of Veterans Affairs

Pittsburgh, PA

Center for Health Equity & Research Promotion - Biostatistics & Informatics Core

- Conduct and disseminate statistical analyses of newly implemented evidence-based practices to clinical leads and investigators of VISN-level initiatives
- Served as a statistician for observational studies of hospice and palliative-care utilization among U.S. Veterans

Data Science Education Consultant

May 2020 - August 2021

University of Pittsburgh, Graduate School of Pharmacy

Pittsburgh, PA

- Developed [open-education resources](#) for students in the university's PharmacoAnalytics program.
- Integrated feedback from industry professionals to ensure resources prepare students for roles as data scientists, analysts, and quantitative researchers.

Data Analyst

April 2018 - May 2020

University of Pittsburgh, School of Medicine

Pittsburgh, PA

- Conducted and presented statistical analyses for grant preparation and submission, national group collaborations, and scientific publications
- Automated data cleaning and quality assurance of genetic, inflammatory biomarker, and functional outcome data using R, Python, and SAS

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Awan N, Kumar RG, Juengst SB, **DiSanto D.**, et al. (2025). "Development of individualized risk assessment models for predicting post-traumatic epilepsy 1 and 2 years after moderate-to-severe traumatic brain injury: A traumatic brain injury model system study." *Epilepsia*, 66: 482-498.
2. Vijapur S, Vaughan LE, **DiSanto D.**, McKernan G, Wagner AK. (2021) "Treelet Transform Analysis to Identify Clusters of Systemic Inflammatory Variance in a Population with Moderate-to-Severe Traumatic Brain Injury." *Brain, Behavior, & Immunity*, 95: 45-60.

3. Millville KM, Awan N, **DiSanto D.**, Kumar RG, Wagner AK. (2021) “Early chronic systemic inflammation and associations with cognitive performance after moderate-to-severe TBI.” *Brain, Behavior, & Immunity – Health*, 11, Article 100185.
4. Awan N*, **DiSanto D.***, et al. (2020) “Evaluating the Cross-Sectional and Longitudinal Relationships Predicting Suicidal Ideation Following Traumatic Brain Injury.” *The Journal of Head Trauma & Rehabilitation*, 36(1): E18-E29.
5. Kumar, R, **DiSanto, D.**, et al. (2020) “Temporal acute serum estradiol and tumor necrosis factor- α associations and risk of death after severe traumatic brain injury.” *Journal of Neurotrauma*, 37(20): 2198-2210.
6. Awan N*, **DiSanto D.***, et al. (2020) “Interrelationships Between Post-TBI Employment and Substance Abuse: A Cross-lagged Structural Equation Modeling Analysis.” *Archives of Physical Medicine and Rehabilitation*, 101(5): 797-806.
7. **DiSanto D.**, et al. (2019). “Employment Stability in the First 5 Years After Moderate-to-Severe Traumatic Brain Injury.” *Archives of Physical Medicine and Rehabilitation*, 100(3): 412-421.

In Submission

8. **DiSanto D.**, et al. “Comparison of B-cell depletion versus natalizumab for treatment of multiple sclerosis: A semi-supervised causal analysis.” [Preprint available at medRxiv](#).

In Preparation

9. **DiSanto, D.** and Lu, J. “Nonparametric, Contextual Evaluation and Assortment Optimization for Decision Agents.”
10. **DiSanto, D.** and Lu, J. “Statistical Inference for Model Class Explanatory Stability.”

* denote joint first-authorship

TEACHING EXPERIENCE

Teaching Fellow (Harvard University)

1. BST 230: Probability I (Fall 2025 & Fall 2026 [*upcoming*])
 - First-year core doctoral course
 - Re-designed lab materials, taught 24 hours of lab lecture, delivered two guest lectures on inequalities and Markov processes, held weekly office hours, graded bi-weekly assignments and exams
2. BST 222: Basics of Statistical Inference (Spring 2025)
 - One semester sequence covering probability and statistical inference from Casella & Berger; targeting master’s students in biostatistics and PhD students in public health (e.g. Global Health, Epidemiology)
 - Taught 15 hours of lab lecture, held weekly office hours, graded bi-weekly assignments, proctored exams
3. BST 210: Applied Regression Analysis (Spring 2024)
 - One semester sequence statistical modeling and introducing students to R; targeted toward master’s students with no prior experience in statistical programming or modeling
 - Taught 15 hours of lab lecture, held weekly office hours, graded bi-weekly assignments, proctored exams

Mentorship & Misc. Instruction

1. Committee Member, Master’s Thesis of Frank Feng. Department of Biostatistics, Harvard T.H. Chan School of Public Health.
 - Thesis: “*Structure-aware prediction of peptide reactivity.*”
2. Instructor, StatStart (Summer 2026)

- Summer program in Harvard’s Department of Biostatistics for high-school students interested in statistics and programming
 - Created materials and delivered lectures covering probability, sampling distributions and the central limit theorem, and network models. Led 3 active-learning lab sessions with interactive R programming exercises
3. Instructor, CELEHS Data Summer Camp (Summer 2026)
 - Summer program for high-school students through CELEHS at Harvard Medical School for students, covering statistical and machine learning concepts
 - Created materials and delivered lecture covering linear models, generalized linear models, and empirical risk minimization (conceptually)
 4. Tutor, Department of Biostatistics PhD Qualifying Exam (Summer 2024 - Summer 2026)
 - 2026 - One of two tutors, held 12 hours of open sessions reviewing relevant content and working through previous years’ qualifying exams
 - 2024 & 2025 - Delivered lectures in asymptotics and convergence of random variables as well as theory and methods in survival analysis

PRESENTATIONS

1. “Expression-Aware Transformers for Serosurveillance and Candidate Peptide Generation”, Joint Statistical Meetings, August 2026.
2. “Non-parametric Evaluation of Contextually Optimal Decision-Aids”, Joint Statistical Meetings, August 2025.
3. “Comparison of B-cell depletion versus natalizumab for treatment of multiple sclerosis: A semi-supervised causal analysis”, American Academy of Neurology Annual Meeting, April 2025.
4. “Peptide-level serosurveillance for vector-borne diseases”, Harvard University Department of Biostatistics HIV Working Group, September 2024. *Lightning Talk*.
5. “Characterizing Genetic Risk Factors of Seizures Following Traumatic Brain Injury”, National Neurotrauma Symposium, 2019.
6. “The interrelationship of post-TBI employment and substance abuse: a cross-lagged structural equation modeling analysis”, National Neurotrauma Symposium, 2019.
7. “Employment Stability in the First 5 Years After Moderate-to-Severe Traumatic Brain Injury”, University of Pittsburgh, Aging Institute Research Day, May 2017, *Top Undergraduate Submission*.

HONORS & AWARDS

Certificate of Distinction in Teaching, Harvard T.H. Chan School of Public Health	Fall 2024 - 2025
GStat Accreditation, American Statistical Association	August 2021
Top MS Student, University of Pittsburgh Department of Biostatistics	April 2021
Pennsylvania Commonwealth Equity Scholarship Recipient	January 2020
Elsie Hillman Fellow, University of Pittsburgh Institute of Politics	Aug 2017 - April 2018
Top Undergraduate Submission, University of Pittsburgh Aging Institute Research Day, May 2017	
Browne Leadership Fellow, University of Pittsburgh School of Social Work	Jan 2016 - Aug 2016
Full Tuition Merit Scholarship Recipient, University of Pittsburgh	Aug 2014 - April 2018

ACADEMIC SERVICE & MEMBERSHIP

Reviewer, Undergraduate Statistics Project Competition	August 2026
Reviewer, ASA Statistics Project Competition	Summer 2026
Volunteer, Data Adventure Day	Summer 2024
Member, Institute of Mathematical Statistics	August 2023 - Present
Member, American Statistical Association	January 2021 - Present
Representative, University of Pittsburgh Mental Health Task Force	January 2020 - May 2020

Ad-hoc reviewer: *Annals of Applied Statistics; International Conference on Machine Learning*

WORKSHOPS

Preparing to Teach Workshop, Joint Statistical Meetings	August 2026
Harnessing Real-World Data and Informatics to Advance Discovery and Care for Chronic Neurological Disorders Workshop	April 2025
Amazon Web Services Immersion Day	May 2024
VERITY Research Bioinformatics Retreat	September 2024
Scholarship Recipient, Summer Institutes in Statistics and Modeling of Infectious Diseases	July 2020